method **is found to reduce** displacement estimation errors by 5 to 50%. With typical clinical data, the hybrid method **can generate more than** 25 strain images per second on commercial hardware; **this is comparable with the alternative approaches considered in this paper**.

5 A ratchet mechanism for amplification in low-frequency mammalian hearing[cc]

Abstract

The sensitivity and frequency selectivity of hearing **result from** tuned amplification by an active process in the mechanoreceptive hair cells. **In most** vertebrates, the active process **stems from** the active motility of hair bundles. The mammalian cochlea **exhibits** an additional form of mechanical activity termed electromotility: its outer hair cells (OHCs) **change** length upon electrical stimulation. The relative contributions of these two mechanisms to the active process in the mammalian inner ear **is the subject of intense current debate. Here, we show that** active hair-bundle motility and electromotility **can** together implement **an efficient mechanism** for amplification that functions like a ratchet: Sound-evoked forces, acting on the basilar membrane, **are transmitted to** the hair bundles, whereas electromotility **decouples** active hair-bundle forces from the basilar membrane. This unidirectional coupling **can extend the hearing range** well below the

resonant frequency of the basilar membrane. **It thereby provides a concept for** low-frequency hearing that **accounts for a variety of unexplained experimental observations** from the cochlear apex, including the shape and phase behavior of apical tuning curves, their lack of significant nonlinearities, and the shape changes of threshold tuning curves of auditory-nerve fibers along the cochlea. The ratchet mechanism **constitutes a general design principle for implementing mechanical amplification in engineering applications**.

6 The osteogenic response of mesenchymal stromal cells to strontium- substituted bioactive glasses[dd]

Abstract

Bioactive glasses **are known to** stimulate bone healing, and the incorporation of strontium **has the potential to increase** their potency. **In this study**, calcium oxide in the 45S5 bioactive glass composition **was** partially (50%, Sr50) or fully (100%, Sr100) **substituted** with strontium oxide on a molar basis. The effects of the substitution on bioactive glass properties **were studied**, including density, solubility, and *in vitro* cytotoxicity. Stimulation of osteogenic differentiation **was investigated using** mesenchymal stromal cells obtained from rat bone marrow. Strontium substitution **resulted in** altered physical properties including increased solubility. Statistically significant reductions in cell viability

were observed with the addition of bioactive glass powders to culture medium. **Specifically**, addition of ⩾ 13.3 mg/ml of 45S5 bioactive glass or Sr50, or ⩾ 6.7 mg/ml of Sr100, **resulted in** significant inhibition. **Real-time PCR analyses detected** the upregulation of genes associated with osteoblastic differentiation in the presence of all bioactive glass compositions. Some genes, including *Alpl* and *Bglap*, **were further stimulated in the presence of** Sr50 and Sr100. **It was concluded that** strontium-substituted bioactive glasses **promoted** osteogenesis in a differentiating bone cell culture model and, therefore, **have considerable potential for use as improved** bioactive glasses for bone tissue Regeneration.

7 An investigation into reinforced and functionally graded lattice structures[ee]

Abstract

Lattice structures **are regarded as excellent candidates for use in** lightweight energy-absorbing applications, such as crash protection. **In this paper we investigate** the crushing behaviour, mechanical properties and energy absorption of lattices made by an additive manufacturing process. Two types of lattice **were examined:** body-centred-cubic (BCC) and a reinforced variant called BCC_z. The lattices **were subject to** compressive loads in two orthogonal directions, **allowing** an assessment of their mechanical anisotropy to be made. **We**

also examined functionally graded versions of these lattices, which featured a density gradient along one direction. The graded structures **exhibited distinct** crushing behaviour, with a sequential collapse of cellular layers preceding full densification. For the BCC_z lattice, the graded structures **were able to** absorb around 114% more energy per unit volume than their non-graded counterparts before full densification, $1371 \pm 9\,kJ/m^3$ versus $640 \pm 10\,kJ/m^3$. **This highlights the strong potential for functionally graded lattices to be used in energyabsorbing applications**. Finally, **we determined** several of the Gibson–Ashby coefficients relating the mechanical properties of lattice structures to their density; **these are crucial** in establishing the constitutive models required for effective lattice design. These results improve the current understanding of additively manufactured lattices **and will enable the design of sophisticated, functional, lightweight components in the future**.

8 A transient vanadium flow battery model incorporating vanadium crossover and water transport through the membrane[ff]

Abstract

This paper presents a 2-D transient, isothermal model of a vanadium redox flow battery **that can predict** the species crossover and related capacity loss during operation. **The**

model incorporates the species transport across the membrane due to convection, diffusion, and migration, **and accounts for** the transfer of water between the half-cells to capture the change in electrolyte volume. **The model also accounts for** the side reactions and associated changes in species concentration in each half-cell due to vanadium crossover. A set of boundary conditions based on the conservations of flux and current **are incorporated** at the electrolyte | membrane interfaces **to account for** the steep gradients in concentration and potential at these interfaces. **In addition, the present model further improves upon the accuracy of existing models by incorporating a more complete version** of the Nernst equation, which **enables accurate prediction** of the cell potential **without the use of** a fitting voltage. A direct **comparison of the model predictions with experimental data shows that the model accurately predicts** the measured voltage of a single charge/discharge cycle with an average error of 1.83%, and estimates the capacity loss of a 45 cycle experiment with an average error of 4.2%.

9 Sensory nerve induced inflammation contributes to heterotopic ossification[r]

Abstract

Heterotopic ossification (HO), or bone formation in soft tissues, **is often the result of** traumatic injury. **Much evidence has linked** the release of BMPs (bone morphogenetic

proteins) upon injury to this process. HO **was once thought to be a rare occurrence, but recent statistics** from the military **suggest that as many as 60%** of traumatic injuries, resulting from bomb blasts, have associated HO. **In this study, we attempt to define the role of** peripheral nerves in this process. Since BMP2 **has been shown previously** to induce release of the neuroinflammatory molecules, substance P (SP) and calcitonin gene related peptide (CGRP), from peripheral, sensory neurons, **we examined** this process in vivo. SP and CGRP **are rapidly expressed** upon delivery of BMP2 **and remain** elevated throughout bone formation. In animals lacking functional sensory neurons ($TRPV1^{-/-}$), BMP2-mediated increases in SP and CGRP **were suppressed** as compared to the normal animals, and HO **was dramatically inhibited** in these deficient mice, **suggesting that** neuroinflammation plays a functional role. Mast cells, known to be recruited by SP and CGRP, **were elevated** after BMP2 induction. These mast cells **were localized** to the nerve structures and **underwent** degranulation. When degranulation **was inhibited using** cromolyn, HO **was again reduced significantly**. Immunohistochemical analysis **revealed** nerves expressing the stem cell markers nanog and Klf4, as well as the osteoblast marker osterix, after BMP2 induction, in mice treated with cromolyn. **The data collectively suggest that** BMP2 **can act directly on** sensory neurons to induce neurogenic inflammation, **resulting in** nerve remodeling and the migration/release of osteogenic and other stem cells

from the nerve. **Further**, blocking this process **significantly reduces** HO, **suggesting that** the stem cell population **contributes to** bone formation.

10 Global anisotropic 3D FWI[gg]

Abstract

Seismic anisotropy **influences** both the kinematics and dynamics of seismic waveforms. **If** anisotropy **is not adequately taken into account** during fullwaveform seismic inversion (FWI), then **inadequacies** in the anisotropy model **are likely to manifest as signifificant error** in the recovered P-wave velocity model. **Conventionally**, anisotropic FWI **uses** either a fixed anisotropy model derived from tomography or such, **or it uses** a local inversion scheme to recover the anisotropy as part of the FWI; **both of these methods can be problematic. In this paper, we show that** global rather than local FWI **can be used** to recover the long-wavelength anisotropy model, and **that this can then be followed by** more-conventional local FWI to recover the detailed model. **We demonstrate this approach on** a full 3D field dataset, **and show that it avoids problems** associated to cross-talk that can bedevil local inversion schemes. **Although our method** provides a global inversion of anisotropy, **it is nonetheless affordable and practical** for 3D field data.

6.2.2 有结构的摘要

11 A finite rate of innovation algorithm for fast and accurate spike detection from two-photon calcium imaging[hh]

Abstract

Objective

Inferring the times of sequences of action potentials (APs) (spike trains) from neurophysiological data is a key problem in computational neuroscience. The detection of APs from two-photon imaging of calcium signals offers certain advantages over traditional electrophysiological approaches, as up to thousands of spatially and immunohistochemically defined neurons can be recorded simultaneously. However, due to noise, dye buffering and the limited sampling rates in common microscopy configurations, accurate detection of APs from calcium time series has proved to be a difficult problem.

Approach

Here we introduce a novel approach to the problem making use of finite rate of innovation (FRI) theory (Vetterli *et al* 2002 IEEE *Trans. Signal Process*: **50** 1417–28). For calcium transients well fit by a single exponential, the problem is reduced to reconstructing a stream of decaying exponentials. Signals made of a combination of exponentially decaying functions with different onset times are a subclass of FRI

signals, for which much theory has recently been developed by the signal processing community.

Main results

Wedemonstrate for the first time the use of FRI theory to retrieve the timing of APs from calcium transient time series. The final algorithm is fast, non-iterative and parallelizable. Spike inference can be performed in real-time for a population of neurons and does not require any training phase or learning to initialize parameters.

Significance

The algorithm has been tested with both real data (obtained by simultaneous electrophysiology and multiphoton imaging of calcium signals in cerebellar Purkinje cell dendrites), and surrogate data, and outperforms several recently proposed methods for spike train inference from calcium imaging data.

12 The *Plasmodium falciparum* cytoplasmic translation apparatus: a promising therapeutic target not yet exploited by clinically approved anti-malarials[ii]

Abstract

Background

The continued spectre of resistance to existing anti-malarials

necessitates the pursuit of novel targets and mechanisms of action for drug development. One class of promising targets consists of the 80S ribosome and its associated components comprising the parasite translational apparatus. Development of translation-targeting therapeutics requires a greater understanding of protein synthesis and its regulation in the malaria parasite. Research in this area has been limited by the lack of appropriate experimental methods, particularly a direct measure of parasite translation.

Methods

An in vitro method directly measuring translation in whole-cell extracts from the malaria parasite *Plasmodium falciparum*, the PfIVT assay, and a historically-utilized indirect measure of translation, S35-radiolabel incorporation, were compared utilizing a large panel of known translation inhibitors as well as anti-malarial drugs.

Results

Here, an extensive pharmacologic assessment of the PfIVT assay is presented, using a wide range of known inhibitors demonstrating its utility for studying activity of both ribosomal and non-ribosomal elements directly involved in translation. Further, the superiority of this assay over a historically utilized indirect measure of translation, S35-radiolabel incorporation, is demonstrated. Additionally, the

PfIVT assay is utilized to investigate a panel of clinically approved anti-malarial drugs, many with unknown or unclear mechanisms of action, and show that none inhibit translation, reaffirming *Plasmodium* translation to be a viable alternative drug target. Within this set, mefloquine is unambiguously found to lack translation inhibition activity, despite having been recently mischaracterized as a ribosomal inhibitor.

Conclusions

This work exploits a direct and reproducible assay for measuring *P. falciparum* translation, demonstrating its value in the continued study of protein synthesis in malaria and its inhibition as a drug target.

6.2.3　包含重要性说明和/或亮点的摘要

请注意，重要性说明和亮点并不总是出现在论文的PDF中，因此应将它们视为摘要的附件而不是补充。使用你的逆向工程技术，描述重要性说明和亮点中的句子功能。模板标识符以**粗体**显示。

13 The oxidative corrosion of carbide inclusions at the surface of uranium metal during exposure to water vapour[d]

Abstract

The reaction between uranium and water vapour **has been well investigated, however discrepancies exist** between the described kinetic laws, pressure dependence of the reaction rate constant and activation energies. **Here this problem is looked at by examining** the influence of impurities in the form of carbide inclusions on the reaction. Samples of uranium containing 600 ppm carbon **were analysed** during and after exposure to water vapour at 19 mbar pressure, in an environmental scanning electron microscope (ESEM) system. After water exposure, samples **were analysed using** secondary ion mass spectrometry (SIMS), focused ion beam (FIB) imaging and sectioning and transmission electron microscopy (TEM) with X-ray diffraction (micro-XRD). **The results of the current study indicate that** carbide particles on the surface of uranium **readily react** with water vapour to form voluminous $UO_3 \cdot xH_2O$ growths **at rates significantly faster than** that of the metal. **The observation may also have implications for previous experimental studies** of uraniumwater interactions, where the presence of differing levels of undetected carbide **may partly account for** the discrepancies observed between datasets.

Highlights

► **High resolution imagery** (FIB, SEM and SIMS) of carbide inclusions in uranium metal. ► **Real time images** following the reaction of the carbide inclusions with water vapour. ► **Shown** preferential consumption of carbide over that of the bulk metal. ► Quantity of impurities in the metal **therefore seriously influence** reaction rate. ► Metal purity **must be considered when** storing uranium in air or moist conditions.

14 Amazonian functional diversity from forest canopy chemical assembly°

Significance

Canopy trees are **keystone organisms** that create habitat for **an enormous array of** flora and fauna and **dominate** carbon storage in tropical forests. Determining the functional diversity of tree canopies **is, therefore, critical to understanding** how tropical forests are assembled and predicting ecosystem responses to environmental change. Across the megadiverse Andes-to-Amazon corridor of Peru, **we discovered** a large-scale nested pattern of canopy chemical assembly among thousands of trees. This nested geographic and phylogenetic pattern within and among forest communities **provides a different perspective on current and future** alterations to the functioning of western Amazonian forests resulting from land use and climate change.

Abstract

Patterns of tropical forest functional diversity **express** processes of ecological assembly at multiple geographic scales and **aid in predicting** ecological responses to environmental change. Tree canopy chemistry **underpins** forest functional diversity, **but** the interactive role of phylogeny and environment in determining the chemical traits of tropical trees **is poorly known. Collecting and analyzing** foliage in 2,420 canopy tree species across 19 forests in the western Amazon, **we discovered** (*i*) systematic, community-scale shifts in average canopy chemical traits along gradients of elevation and soil fertility; (*ii*) strong phylogenetic partitioning of structural and defense chemicals within communities independent of variation in environmental conditions; and (*iii*) strong environmental control on foliar phosphorus and calcium, the two rock-derived elements limiting CO_2 uptake in tropical forests. **These findings indicate that** the chemical diversity of western Amazonian forests **occurs** in a regionally nested mosaic driven by long-term chemical trait adjustment of communities to large-scale environmental filters, particularly soils and climate, **and is supported by** phylogenetic divergence of traits essential to foliar survival under varying environmental conditions. Geographically nested patterns of forest canopy chemical traits **will play a role in determining** the response and functional rearrangement of western Amazonian ecosystems to changing land use and climate.

15 Ultrafast desorption of colloidal particles from fluid interfaces[ij]

Significance

Solid particles **can** replace surfactants to stabilize emulsions and foams. The attachment of particles onto drops and bubbles is **typically considered to be** irreversible because of a large energy barrier for particle detachment — millions of times the thermal energy for microparticles. **Here we demonstrate a method to** promote the detachment of microparticles from bubbles using ultrasound. **We identified** conditions for complete particle removal and recovery in **under a millisecond. Our method** is programmable in time, and does not require any physicochemical modification of the fluids or the interface. **This work addresses the emerging need for** methods to recover interfacial particles from emulsions and foams **in applications ranging from** controlled release to interfacial catalysis and gas storage.

Abstract

The self-assembly of solid particles at fluid–fluid interfaces **is widely exploited** to stabilize emulsions and foams, and in materials synthesis. The self-assembly mechanism **is very robust** owing to the large capillary energy associated with particle adsorption, of the order of millions of times the thermal energy for micrometer-sized colloids. The microstructure of the interfacial colloid monolayer **can also favor** stability, for

instance in the case of particle-stabilized bubbles, which **can be** indefinitely stable against dissolution due to jamming of the colloid monolayer. **As a result, significant challenges arise** when destabilization and particle removal **are a requirement. Here we demonstrate** ultrafast desorption of colloid monolayers from the interface of particle-stabilized bubbles. **We drive** the bubbles into periodic compression–expansion using ultrasound waves, causing significant deformation and microstructural changes in the particle monolayer. **Using** high-speed microscopy **we uncover different** particle expulsion scenarios depending on the mode of bubble deformation, including highly directional patterns of particle release during shape oscillations. **Complete** removal of colloid monolayers from bubbles **is achieved in under a millisecond. Our method should find a broad range of applications**, from nanoparticle recycling in sustainable processes to programmable particle delivery in lab-on-a-chip applications.

6.2.4 图形摘要

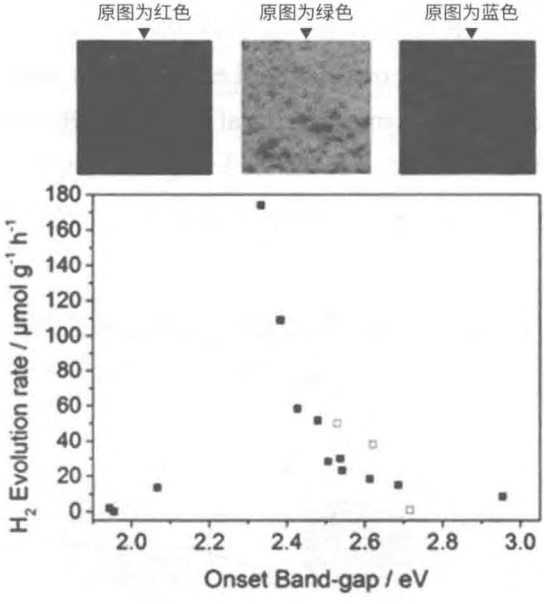
图形摘要(Graphical Abstract, GA)通常设计成单幅图像，旨在让读者立即了解论文的关键信息。GA应该明智地使用颜色，并通过最大限度地减少杂乱和不必要的元素来在视觉上吸引注意力。在一些情况下，图像是从论文中复制出来的；在其他情况下，也会专门为GA设计图像。

GA一开始就表达了主要观点，这提高了读者对信息的理

解和接受程度。此外，即使在读者移至另一篇文章之后，提供清晰且令人难忘的图像也能增强对该信息的记忆力。请注意，一些GA还包含一个简短的文本。

使用你的逆向工程技术，为下面的GA构建一个模板，并将其与第6.3节中的通用模板进行比较。模板标识符以**粗体**显示。

16 Tunable organic photocatalysts for visible-light-driven hydrogen evolution[kk]

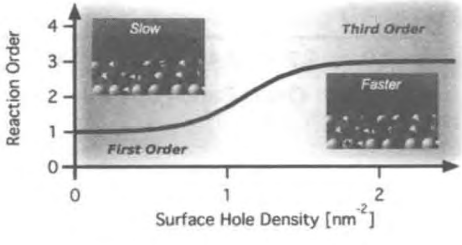


Abstract

Photocatalytic hydrogen production from water **offers an**

abundant, clean fuel source, **but it is challenging to produce** photocatalysts that use the solar spectrum **effectively**. Many hydrogen-evolving photocatalysts **are** active in the ultraviolet range, **but** ultraviolet light **accounts for only** 3% of the energy available in the solar spectrum at ground level. Solid-state crystalline photocatalysts have light absorption profiles that are a discrete function of their crystalline phase and that are not always tunable. **Here, we prepare** a series of amorphous, microporous organic polymers **with exquisite synthetic** control over the optical gap in the range 1.94–2.95 eV. Specific monomer compositions give polymers that are **robust and effective** photocatalysts for the evolution of hydrogen from water in the presence of a sacrificial electron donor, **without the apparent need for** an added metal cocatalyst. **Remarkably, unlike other** organic systems, the best performing polymer is only photoactive under visible rather than ultraviolet irradiation.

17 Rate law analysis of water oxidation on a hematite surface[u]



Abstract

Water oxidation **is a key chemical reaction, central to** both biological photosynthesis and artificial solar fuel synthesis strategies. **Despite recent progress** on the structure of the natural catalytic site, and on inorganic catalyst function, determining the mechanistic details of this multiredox reaction **remains a significant challenge. We report herein** a rate law analysis of the order of water oxidation as a function of surface hole density on a hematite photoanode employing photoinduced absorption spectroscopy. **Our study reveals** a transition from a slow, first order reaction at low accumulated hole density to a faster, third order mechanism once the surface hole density is sufficient to enable the oxidation of nearest neighbor metal atoms. **This study thus provides direct evidence for** the multihole catalysis of water oxidation by hematite, **and demonstrates** the hole accumulation level required to achieve this, **leading to key insights** both for reaction mechanism and strategies to enhance function.

一些期刊会接受或者鼓励作者把亮点或重要性说明和GA放在一起。

18 Influence of reaction conditions on the composition of liquid products from two-stage catalytic hydrothermal processing of lignin[II]

Highlights

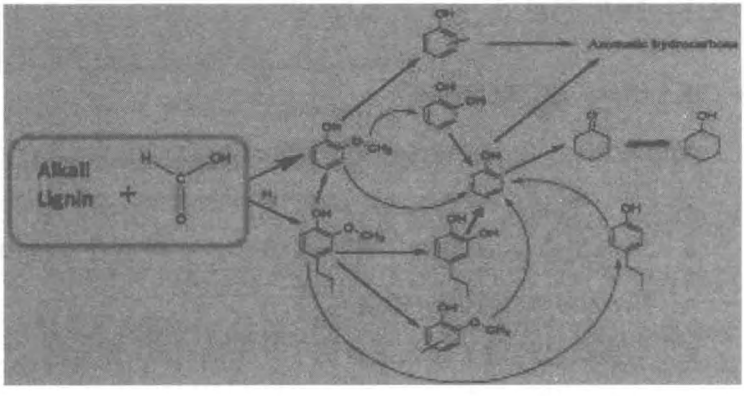
- Liquid products' yields **reached up to** 40 wt% based on lignin feed.
- Liquid products' compositions **differed** with different reaction conditions.
- Formic acid **acted as** both depolymerization agent and source of hydrogen.
- Formic acid **influenced** liquid products' selectivity under the different conditions.
- Pt/Al_2O_3 **was** less selective towards liquid products than formic acid.

Abstract

The influence of reaction conditions on the composition of liquid products during two-stage hydrothermal conversion of alkali lignin **has been investigated** in a batch reactor. Reactions **were carried out** in the presence of formic acid (FA) and Pt/Al_2O_3 catalyst. The two different sets of reaction conditions **involved** alternative reaction times of 1 h and 5 h at 265°C and 350°C, respectively. These **provided different contributions** to reaction severity, **which affected** the compositions of liquid

products. Yields of liquid products **reached up to** 40 wt% (on lignin feed basis) in the presence of FA under the less severe reaction condition. With 5 h reaction time at 350°C, alkylphenols, alkylguaiacols and hydrocarbons **were the dominant** liquid products. However, with 5 h reaction time at 265°C, phenol and methanol **became dominant**. The two-stage hydrothermal process **led to improved** lignin conversion, with the **potential** to manipulate the liquid product range.

Graphical abstract



6.3　通用摘要模板

现在根据下面的通用模板比较和调整你的模板。请注意可以**快速和无误地**表达研究价值的“快乐的单词”☺的潜在位置的数量。

	通用摘要模板
1	标题的重要性/基本事实背景 挑战/问题 论文/研究做了什么(可能包括成就/价值+☺)
2	方法/材料+☺ 结果+☺/和已有结果比较+☺ 含义
3	映射已有的知识+☺ 成就/价值/贡献+☺ 应用+☺

6.3.1 模板组件

▶ 每个组件应该用哪种时态?

和以往一样,你的目标文章提供最好的指导,但一般而言:

- 当提及论文/研究做了什么或者在论文里有什么的时候,大多数作者用一般现在时。
- 当提及已经做了什么或者用了什么的时候,即方法,大多数作者用一般过去时。
- 结果和含义可以用一般过去时或一般现在时。摘要以一种能够立即吸引读者注意力的方式呈现论文的内容,因此,摘要中的结果和含义有时会使用现在时,即使这些结果和含义在论文正文中以过去时表示。
- 成果/价值/贡献可以用现在完成时(have+ed)或一般现在时来表达。

我怎样才知道我需要说明主题的重要性还是提供背景信息?

这在一定程度上取决于期刊的读者群和研究的具体性。然而，即使在比较有技术性的领域内，你也应该考虑研究领域范围之外的潜在跨学科读者的需求。如果需要背景信息来理解摘要本身，请用尽可能少的词来组合和总结相关要点。

我可以在摘要包含引文吗? 例如为了支持背景的引文。

研究背景可能对识别问题或挑战有帮助，但摘要中很少包含实际的引用。但是，如果你的文章直接源于已发表的论文，或者是一个具体工作或理论的重大提高，或是与之不一致，请引用相关论文。

我应该提供方法和结论的多少细节?

关于方法，如果读者在不知道你是否使用模拟、模型或领域数据的情况下无法决定是否阅读该论文，请包含那些方法信息。如果你的研究的**价值**在于你进行了一系列实验而不是单个案例研究，请包含实验信息。

方法的关键方面和关键结果有时与数字细节一起提供：

- The front-end was fabricated in the AMS 0.35 μm technology and occupies an area of 1.35 mm^2.
- Extensive measured results, both electrical and physiological from human subjects are reported, demonstrating an estimated SNR of 39 dB and ability to detect 2% changes in SpO2,…

- Specifically, addition of ⩾ 13.3 mg/ml of 45S5 bioactive glass or Sr50, or ⩾ 6.7 mg/ml of Sr100, resulted in significant inhibition.
- ...the graded structures were able to absorb around 114% more energy per unit volume than their non-graded counterparts before full densification, 1371 ± 9 kJ/m^3 versus 640 ± 10 kJ/m^3.

注：为数字添加语言（例如，**only** 38% 或 **as high as** 38%）可确保数字在这个关键阶段不会被误解。

在其他摘要中，方法和/或结果仅使用语言呈现：

- Stimulation of osteogenic differentiation was investigated using mesenchymal stromal cells obtained from rat bone marrow.
- Here we demonstrate a method to promote the detachment of microparticles from bubbles using ultrasound...
- ...BMP2-mediated increases in SP and CGRP were suppressed as compared to the normal animals, and HO was dramatically inhibited in these deficient mice.
- ...the FWI velocity model reveals a sharper and more intense low-velocity region associated with the gas cloud in which low-velocity fingers match the location of gas-filled faults...

作者在研究本身上已经花费了大量时间，更不用说在撰写论文上花费的时间了，这之后作者往往倾向于将摘要视为所

有工作的概要，并包含许多与方法或结果无关的细节。这导致了一个不成型的、参差不齐的摘要，在这样的摘要中，读者的任务是确定方法的关键方面或关键的结果，这一任务不太可能成功。

我应该要多强有力地陈述结果的影响？

在摘要中所说的应该与你在论文中报告的内容一致。摘要需要令人印象深刻和吸引人，加上作者需要快速发表论文，这可能导致摘要包含骛远的陈述或结论，或者不完全被研究中获得的数据支持的陈述或结论。这一问题将在同行审稿期间暴露，并可能延长审稿过程。如果你的研究是整个突破过程中的起始阶段，或者它的价值在于你的数据可能的或潜在的含义，那么要用情态动词（could/might/may）或像possible/potential这样的单词来表达这一点。在某些情况下，这些影响在摘要中被强有力地表述，而在论文中包含这些含义的可能的局限性的讨论。然而，摘要中陈述的含义不能强有力到误导或与数据本身不一致。

在有字数限制的情况下，我怎样才能使摘要保持流畅？

摘要必须限制在一定的字数内，这有时意味着它要以牺牲叙述的流畅性为代价去包含有价值的数据或信息。然而，叙述性“包装”越少，数据或信息变得越不连贯。

尽可能在限制字数内最多地保留信息的同时使摘要流畅：

1. 复习**第1.5.2节中的句子和信息的连接**，看看本单元中摘要的句子和目标文章中的句子是如何连接在一起的。读者是否容易协商句子之间的差距？作者如何帮助读者从一个句子移到下一个句子？是否有一些地方缺乏明确的句子连接会造成真正的歧义？你怎样能解决这个问题？
2. 通过合并句子功能从受限的字数中得到最大值，例如：

- 论文做了什么+成果："In this paper, we present a novel estimation method that combines the strengths of quality-guided tracking, multi-level correlation, and phase-zero search to achieve high levels of accuracy and robustness."
- 成果+空白："It thereby provides a concept for low-frequency hearing that accounts for a variety of unexplained experimental observations from the cochlear apex, including..."
- 方法+结果："Specifically, addition of ≥13.3mg/ml of 45S5 bioactive glass or Sr50, or ≥ 6.7mg/ml of Sr100, resulted in significant inhibition."
- 成果+空白："In addition, the present model further improves upon the accuracy of existing models by incorporating a more complete version of the Nernst equation, which enables accurate prediction of the cell

potential without the use of a fitting voltage."

- 成果+含义："In animals lacking functional sensory neurons ($TRPV1^{-/-}$), BMP2-mediated increases in SP and CGRP were suppressed as compared to the normal animals, and HO was dramatically inhibited in these deficient mice, suggesting that neuroinflammation plays a functional role."

6.4　语言

摘要所需的所有语言都可以在前面的单元中找到。然而，由于摘要需要被广大读者快速理解，所以它往往包含简单的、传统的语言，下面的节选反映了这一点。根据目标文章的摘要部分补充该列表。

主题的重要性/必要的事实背景

is assumed to	a number of studies
is based on	central
is determined by	common
is fundamental to	currently
is influenced by	dominant
is known to	emerging
is related to	frequently

is regarded as	generally
it has recently been shown that	increasing
it is known that	key
it is widely accepted that	many
	often
	popular
注：一般现在时中的动词也传达基于事实的背景	recent
	typically
	widely
	worldwide

下面是本单元摘要中应用的一些例子：

- The reservoir **lies at** the crest of a high-velocity antiformal chalk section, overlain by about 3000 m of clastics within which a low-velocity gas cloud **produces** a seismic obscured area.
- The sensitivity and frequency selectivity of hearing **result from** tuned amplification by an active process in the mechanoreceptive hair cells.
- Salt sequences **form** an integral part of **many** sedimentary basins worldwide.
- Axial displacement estimation **is fundamental to** many freehand quasistatic ultrasonic strain imaging systems.
- Canopy trees are keystone organisms that create habitat for an **enormous** array of flora and fauna and **dominate** carbon storage in tropical forests.

- The self-assembly of solid particles at fluid–fluid interfaces **is widely exploited** to stabilize emulsions and foams, and in materials synthesis.

挑战/问题

(an) alternative approach	inaccurate
(a) key problem	inadequate
a need for	inconvenient
challenge	limited
complicated	little (work)
critical	not able to
debate	previously
desirable	problem
difficulty	require
disadvantage	risk
drawback	time-consuming
essential	unsuccessful
expensive	until now
impractical	

下面是本单元摘要中应用的一些例子：

- **Despite this, little work has so far been undertaken** to understand magma-salt interactions within the subsurface,…

- The relative contributions of these two mechanisms to the active process in the mammalian inner ear is **the subject of intense current debate**.
- If anisotropy is **not adequately taken into account** during full-waveform seismic inversion (FWI), then inadequacies in the anisotropy model are likely to manifest as significant error…
- …accurate detection of APs from calcium time series **has proved to be a difficult problem**.
- Inferring the times of sequences of action potentials (APs) (spike trains) from neurophysiological data **is a key problem** in computational neuroscience.

论文/研究做了什么 (有时用被动语态)

address	define	extend	propose
analyse	describe	identify	provide
attempt to	discuss	include	report
compare	emphasise	introduce	review
consider	enable	investigate	show
	examine	present	

下面是本单元摘要中应用的一些例子：

- This paper **presents** the implementation of the first fully integrated pulse oximeter front-end with a power

consumption lower than 1 mW.

- The paper **includes** a full description of the hybrid method, in vivo examples to illustrate the method's clinical relevance, and finite element simulations to assess its accuracy.
- In this paper **we investigate** the crushing behaviour, mechanical properties and energy absorption of lattices made by an additive manufacturing process.
- In this study, we **attempt to define** the role of peripheral nerves.
- We **report** herein a rate law analysis of the order of water oxidation as a function of surface hole density on a hematite photoanode employing photoinduced absorption spectroscopy.

方法/材料

Instead of	are/were examined
Rather than	are/were formulated
Unlike	are/were measured
	are/were modelled
are/were analysed	are/were performed
are/were applied	are/were recorded
are/were assembled	are/were selected
are/were calculated	are/were studied
are/were constructed	are/were treated
are/were evaluated	are/were used

下面是本单元摘要中应用的一些例子：

- At each iteration, **we selected** only a subset of sources, **using** a different subset at each iteration…
- In this study, calcium oxide in the 45S5 bioactive glass composition **was** partially (50%, Sr50) or fully (100%, Sr100) **substituted** with strontium oxide on a molar basis.
- Two types of lattice **were examined**: body-centred-cubic (BCC) and a reinforced variant called BCC_z.
- Samples of uranium containing 600 ppm carbon **were analysed** during and after exposure to water vapour at 19 mbar pressure, in an environmental scanning electron microscope (ESEM) system.
- The two different sets of reaction conditions **involved** alternative reaction times of 1 h and 5 h at 265°C and 350°C, respectively.

结　果

cause	are/was achieved
decrease	are/was found
demonstrate	are/was identical
exhibit	are/was identified
increase	are/was observed
occur	are/was obtained
produce	are/was present

reach result in reveal yield	are/was unaffected (by)

下面是本单元摘要中应用的一些例子：

- Extensive measured results, both electrical and physiological from human subjects are reported, **demonstrating** an estimated SNR of 39 dB…
- Deeper in the section, the FWI velocity model **reveals** a sharper and more intense low-velocity region.
- Statistically significant reductions in cell viability **were observed** with the addition of bioactive glass powders to culture medium.
- The graded structures **exhibited** distinct crushing behaviour, with a sequential collapse of cellular layers preceding full densification.
- Yields of liquid products **reached** up to 40 wt% (on lignin feed basis) in the presence of FA under the less severe reaction condition…

成果/价值/贡献

accurate affordable	improved new	able to achieve

better	novel	allow
comparable	powerful	confirm
consistent	practical	enable
cost-effective	reliable	enhance
dramatic	robust	ensure
effective	significant	guarantee
efficient	similar	outperform
exact	simple	simplify
fast	suitable	solve
(the) first	superior	validate
good		verify

大多数摘要包含至少一个使得研究的成果/价值/贡献清晰的句子。以下是前五篇摘要中的一些示例，继续识别摘要的其余部分和目标文章中具有与示例相似功能的句子。

- This paper presents the implementation of **the first fully integrated pulse oximeter front-end with a power consumption lower than 1 mW**.
- **We have developed and implemented a robust and practical scheme** for anisotropic 3D acoustic full-waveform inversion (FWI).
- Within this paper, we detail **the first direct description** of relationships and textures that are developed during intrusive igneous-salt interaction…

- In this paper, **we present a novel estimation method** that combines the strengths of quality-guided tracking, multi-level correlation, and phase-zero search to achieve **high levels of accuracy and robustness**.
- Here, we show that active hair-bundle motility and electromotility can together implement **an efficient mechanism** for amplification that functions like a ratchet:…

影　响

appear to	potentially
indicate that	seem to
may/might/could	suggest that
possible	we conclude that

下面是本单元摘要中应用的一些例子：

- **We suggest that** heating and transformation of carnallite and other hydrous salts into viscous fluidal horizons during igneous intrusion within a regional salt sequence **may act as a possible trigger for contemporaneous** halokinesis.
- **It was concluded that** strontium-substituted bioactive glasses promoted osteogenesis in a differentiating bone cell culture model.
- The data collectively **suggest that** BMP2 can act directly on sensory neurons to induce neurogenic inflammation,

resulting in nerve remodeling and the migration/release of osteogenic and other stem cells from the nerve. Further, blocking this process significantly reduces HO, **suggesting that** the stem cell population contributes to bone formation.

- SND2 **seems to** occupy a subordinate but central tier in the secondary cell wall transcriptional network.
- These findings **indicate that** the chemical diversity of western Amazonian forests occurs in a regionally nested mosaic driven by long-term chemical trait adjustment of communities to large-scale environmental filters…

应　用

apply	potential
employ	relevant for/in
enable	suitable for/in
implement	use
	wide range of

下面是本单元摘要中应用的一些例子：

- The ratchet mechanism constitutes a general design principle **for implementing** mechanical amplification **in engineering applications**.
- The workflows, inversion strategy, and algorithms that we have used have **broad application** to invert a **wide range**

of analogous data sets.

- These results improve the current understanding of additively manufactured lattices and will **enable** the design of sophisticated, functional, lightweight components in the future.
- It was concluded that strontium-substituted bioactive glasses promoted osteogenesis in a differentiating bone cell culture model and, therefore, have considerable **potential for use as** improved bioactive glasses for bone tissue regeneration.
- Geographically nested patterns of forest canopy chemical traits **will play a role in** determining the response and functional rearrangement of western Amazonian ecosystems to changing land use and climate.
- **Our method should find a broad range of applications,** from nanoparticle recycling in sustainable processes to programmable particle delivery in lab-on-a-chip applications.

6.5　摘要总结练习

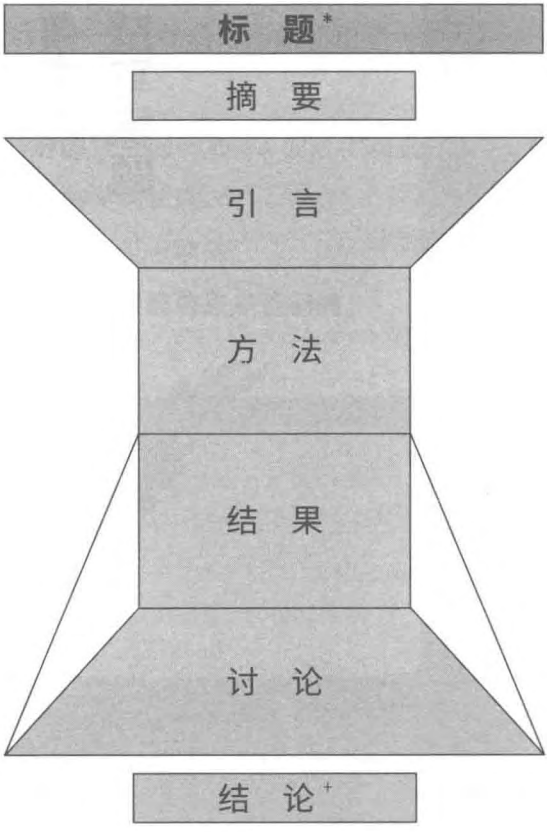
分析目标文章的摘要以确定以下所有内容：

- 结构对应或偏离第6.3节的通用摘要模板的程度。
- 重复/回顾方法和结果在摘要的比例。
- 第一个句子的功能。
- 指明研究价值/成就的“快乐”语言的数量。
- 是否有降低风险的语言，包括像may 一样的情态动词。

- 研究的主要成就是如何被优先考虑的。
- 每一个句子中的动词时态是什么，并且为什么使用这个时态。
- 动词时态或被动语态的使用是否造成任何“所有权”歧义。
- 是否因参考词如 this/it 而产生歧义。
- 句子的平均长度。
- 哪些缩写是可取的。
- **该研究的关键信息是否明确**。

第七单元
撰写标题





* 为简单起见，所有标题都在大写方面进行了标准化。

\+ 有些期刊在这一部分使用结论一词的单数形式，有些使用复数形式，但这似乎并不能反映所得结论的数量。

研究论文的标题经历了根本性的变化，并且正在继续发展以满足读者的需求。在以往这几年，研究人员通过精心策划的数量有限的期刊论文复印件来了解他们领域的发展。现在情况已经不再是这样了。这种转变的主要驱动力是研究的在线阅读，这对标题的目标、内容和结构产生了深远的影响。互联网阅读意味着阅读标题的人比阅读论文任何其他部分的人要多，而搜索引擎的优化意味着许多人一旦找到了相关研究后就不会再滚动超过标题后的几页。浏览标题现在是大多数学者的一项持续活动，因此标题变得越来越重要。

其他因素也改变了标题的写作方式和它们面向的群体。例如，如今有许多高度专业化的研究微群体，大量的知识在其群体内部被认为是理所当然的，并且针对这些微群体的研究论文往往在标题中包含高度专业化的内容。相比之下，其他人处在需要让跨学科读者可以了解他们的研究工作的领域，这促使这些研究人员在创建标题时需要优先考虑广泛的影响和包容性而不是内容。

标题将会永远存在于论文中，并始终作为潜在读者的切入点。一个好的标题将会吸引读者，重要的是，会吸引所有合适的读者。反之亦然：如果标题糟糕，论文可就不会被目标读者读到。在你投稿之前，先在搜索引擎中输入你的标题和关键字列表，检查一下出现了哪些文章。他们的目标读者群是否与你的相同？那是你的目标读者吗？

尽管标题很重要，但关于什么是“好”标题的建议有时过

于笼统而实用性不高，在某些情况下可能与实际操作相矛盾。你可能会被告知你的标题应该是“提供信息的”，但这些信息到底应该是什么并不容易确定。你可能会被建议要让你的标题“尽可能短”，但是当你查看目标期刊中类似的近期论文的标题时，它们可能根本不短——无论如何，“尽可能短”到底有多短？此外，鉴于发表研究成果的变化速度越来越快，即使是有用的建议也可能很快就会过时。缺乏有效性的建议意味着作者有时只是简单地提交带有原始工作标题的文章并希望结果会是好的，但在竞争激烈的全球研究背景下，这不是一个好策略。

▶ 使用工作标题有什么问题呢？

工作标题是在研究活动之前和期间存在于研究人员头脑中的研究主题，至关重要的是，它是在获得结果之前就存在的。它没有告诉读者他们可以从阅读文章中获得什么，而是倾向于反映研究领域，充其量是反映研究活动。而研究活动可能仅仅指**对某件事情的研究**，但是不但**研究活动**本身是很模糊的，而且该研究的产品，也即价值，并不是研究活动本身，而是研究的结果。

此外，工作标题有时只是命名你正在研究的事物，例如，**管网中的不稳定流**（**Unsteady flow in pip networks**），而该研究的产品或结果，也就是论文标题，可能是**不稳定流管网的水力分析**（**A hydraulic analysis of unsteady flow in pipe network**）。即使工作标题非常具体，例如**映射肺中单细胞基因表达**

（**Mapping single-cell gene expression in the lung**），这仍然是研究活动，它不是最好的或提供最多信息的标题。这篇论文是在描述映射单细胞基因表达的新过程，指明映射单细胞基因表达的挑战，还是简单地综述已有文献？如果该研究的结果以诸如《**LungGENS：一种基于网络的映射发育中肺中单细胞基因表达的工具**》（"**LungGENS: a web-based tool for mapping single-cell gene expression in the developing lung**"）这样的标题来表达，将会吸引读者，因为他们确切地知道用他们非常宝贵的时间阅读这篇论文会得到什么。

作者只能在研究已经完成，并且研究结果已经根据当前的文献和知识进行记录和分析后，才能真正确定他们的贡献并决定合适的标题。这个决定有时是在非常接近投稿时间的时候才做出的，最终的标题往往与原来的工作标题几乎没有相似之处。

我不能只用一句话概述我的研究结果来生成标题吗？

句子标题在一些研究领域是可以接受的，并且越来越倾向于将它们用作小节的摘要标题。句子标题有一个语法主语、一个动词，通常还有一个宾语："Gamma-range synchronization of fast-spiking interveurons **enhances** detection of tactile stimuli." 概述最重要研究的句子标题在某些期刊中是可以接受的，但并不常见——与往常一样，请检查你的目标期刊。当研究本身强有力的时候可以使用句子标题，这些句子简单地陈述了关键研究

的本身，通常用一般现在时态。鉴于发表研究成果对速度的需要，在某些情况下，即使研究结果的含义尚未完全确定，期刊编辑也会接受句子标题："Mechanically modulated cartilage growth **may** regulate joint surface morphogenesis."

▶ 那么我应该使用什么策略来生成标题？

如上所述，标题正在改变以适应读者访问研究的方式，因此最好的策略是首先对你希望发表的期刊中的当前标题示例和/或处理类似主题的研究文章的标题进行逆向工程。**使用以下6点来分析目标期刊中最近可比较的标题**。

7.1 检查平均长度

这是一个合理的起点，因为平均字数为你的目标期刊标题的复杂程度提供了一些粗略的指导。例如，在很多期刊中，标题的平均字数为12，但平均字数总是隐藏着可接受的变动。实际数字可能在很大程度上取决于具体主题和研究类型。此外，虽然知道平均字数是一个很好的起点，但该数字很可能会随着研究文章在网页头版的未来变化而变化。一些网页头版现在包括诸如**重要性声明**（**Significance Statement**）或**研究背景**（**Research in Context**）之类的部分。这些组成部分的数量和范围在不同期刊之间并不是一成不变或一致的，最终可能会影响标题的功能，进而影响其中字数的中位数长度。

7.2　使用缩写

使用缩写来缩小标题是很诱人的，但不应随便决定使用缩写。在你进行研究时，你大部分时间将与分享和研究相关概念和术语知识的同事一起度过，以至于你的大部分交流都依赖于这种隐含的共同理解。到你写论文的标题时，你已经和同事们频繁地使用了一组共享的缩写，以至于完整的形式似乎是多余的。这往往会使你看不到这样一个现实：阅读你作品的人可能并不确切知道你的缩写指的是什么——他们可能只是发现标题中使用的缩写令人反感，然后向下滚动到下一篇文章。

此外，尽管标题中包含缩写在高度特定的微型研究群体中是可以接受的，但这些缩写对于局外人来说是完全无法理解的，而今天的局外人很可能成为明天的跨学科读者。例如，一篇关于使用机器学习识别疟疾疫苗候选者的研究论文的潜在未来读者最终可能包括广泛的利益相关者和读者，如癌症研究人员、公共卫生政策制定者、临床医生和制药业投资者，因此决定是否在标题中使用缩写取决于正确估计当前和未来读者共享的知识水平。

7.3　将标题关键字与关键字列表进行比较

关键字是营销工具，期刊通常会要求文章提供简短的**关键字列表**以供显示在首页。该列表通常包含增强索引的词汇或短

语，从而增强文章的可搜索性。关键字列表是对标题的一个有价值的补充，因此创建一个简单重复标题中的关键字的关键字列表并不是一种富有成效的策略。

将关键字列表与标题中的关键字进行比较对提供信息是非常有用的。在某些情况下，关键字列表比标题中的关键字更宽泛，表明读者群可以比标题所建议的更广泛。例如，一篇标题为《一个关于局部可再生能源、食物、供给和木材生产的检验框架》（"A framework for reviewing the trade-offs between renewable energy, food, feed and wood production at a local level"）的文章的关键字列表包括**生态系统**（Ecosystem）和**风景**（Landscape）这样的词，以吸引尽可能广泛的受众。另一方面，题目也许会聚焦于研究的更广泛的影响和贡献，而采用技术性较强的关键字可以吸引该领域的专业研究人员。例如，一篇标题为《大型火成岩范围和有机碳掩埋：早白垩世期间对全球温度和大陆风化的控制》（"Large igneous provinces and organic carbon burial: controls on global temperature and continental weathering during the Early Cretaceous"）的论文的关键字列表包括**箭石**（Belemnites）、**海洋缺氧事件**（Oceanic Anoxi Events）和**晚期寒流**（Late Apitan cold snap）这样的关键字。

将内容转移到关键字列表可以减少题目的过度拥挤，这是关键字列表的另一个有用功能。为了优化搜索引擎，在标题中包含所有潜在关键字可能会使标题中的名词过量。这往往会分散关键信息，因为读者无法决定中心焦点在哪里。例如，诸如

"Simultaneous voltage and calcium mapping of genetically purified human induced pluripotent stem cell-derived cardiac myocyte monolayers"(《基因纯化的人诱导多能干细胞驱动的心肌细胞单层的同时电压和钙检测》)之类的标题并不能使读者在标题滚动时快速识别研究的中心焦点。

7.4　检查标题的语法

这听起来很明白易懂,但标题对读者来说必须是有意义的。标题中的名词和复合名词越多,就越难传达清晰的信息。复合名词尤其成问题。它们之所以吸引人,是因为它们压缩了含义,并且可以给人一种新的或修改过的概念的"所有权"印象,但是随着它们变长,标题的整体语法和每个名词之间的关系变得更难控制。复合名词右侧的名词在语法上是唯一的"实际"名词;它左边的名词具有与形容词相同的功能,因为它们修饰名词。随着复合名词变长,这些名词之间的关系变得难以阐明:

- an **oil can** = 一个装油的罐(**can**)
- an **oil can opener** = 一个装油的罐子的开罐器(**opener**)
- an **oil can opener repair technician** = 一个修理装油的罐子的开罐器的技术(**technician**)
- an **oil can opener repair technician training**